

Fault Diagnosis of Rotating Machines Using Vibration Signals and Deep Learning

A Technical Report for Project Submission

Proposed Hybrid Deep Learning Architecture
for Vibration-Based Condition Monitoring

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1. Abstract

Rotating machines—including motors, pumps, turbines, compressors, and gearboxes—are fundamental components of modern industrial infrastructure. Their reliable operation is critical to manufacturing, power generation, transportation, and numerous other sectors. Unexpected failures of these machines can lead to costly unplanned downtime, safety hazards, and significant financial losses. Consequently, early and accurate fault diagnosis is essential for implementing effective predictive-maintenance strategies.

Vibration signals collected from accelerometers mounted on rotating machinery carry rich information about the machine's health condition. Traditional approaches to vibration-based fault diagnosis rely on manual feature extraction in the time domain, frequency domain, or both—a process that requires substantial domain expertise and is often sensitive to operating conditions. In recent years, deep learning has emerged as a powerful alternative, offering the ability to automatically learn discriminative features directly from raw or transformed sensor data.

This report presents a proposed hybrid deep learning architecture for vibration-based fault diagnosis of rotating machines. The architecture employs a dual-branch feature-extraction strategy: a one-dimensional convolutional neural network (1D CNN) processes raw time-domain vibration windows, while a two-dimensional convolutional neural network (2D CNN) processes spectrograms generated via the Short-Time Fourier Transform (STFT). The features learned by both branches are fused through concatenation and subsequently processed by a Bidirectional Long Short-Term Memory (BiLSTM) network to capture temporal dependencies. A fully connected layer followed by a Softmax classifier produces the final fault-class predictions. Additionally, an optional fuzzy logic module is proposed for post-classification risk and severity assessment, combining deep-learning fault probabilities with statistical signal features to generate interpretable risk levels.

This architecture is inspired by and builds upon existing research in vibration-based condition monitoring, convolutional neural networks, time-frequency analysis, and recurrent neural networks. The contribution of this work lies in the systematic integration of these established techniques into a coherent, end-to-end diagnostic pipeline.

2. Introduction

2.1 Rotating Machines in Industry

Rotating machines are ubiquitous in industrial settings. Electric motors, generators, pumps, fans, compressors, turbines, and gearboxes constitute the backbone of modern manufacturing and energy systems. These machines operate under varying loads, speeds, and environmental conditions, and their continuous, reliable functioning is essential for maintaining production efficiency and ensuring workplace safety.

2.2 Common Faults in Rotating Machines

Despite careful design and manufacturing, rotating machines are susceptible to a range of faults that develop over time due to wear, fatigue, contamination, and operational stress. The most common fault categories include:

- **Bearing faults:** Defects in the inner race, outer race, rolling elements, or cage of bearings. Bearing faults account for a significant proportion of all rotating-machine failures.
- **Unbalance:** An uneven distribution of mass around the axis of rotation, causing excessive vibration at the rotational frequency.
- **Misalignment:** Angular or parallel offset between coupled shafts, generating characteristic vibration patterns at harmonics of the rotational speed.
- **Gear faults:** Tooth wear, pitting, cracking, or breakage in geared systems, producing distinctive vibration signatures related to the gear-mesh frequency.
- **Looseness:** Structural or mechanical looseness at mounting points or bearings, leading to broadband vibration increases.

2.3 Importance of Predictive Maintenance

Industrial maintenance strategies have evolved from reactive (run-to-failure) and preventive (time-based) approaches toward predictive (condition-based) maintenance. Predictive maintenance leverages real-time or periodic condition-monitoring data to detect incipient faults before they progress to catastrophic failure. The benefits include:

- Reduction of unplanned downtime
- Extension of machine service life
- Optimisation of maintenance scheduling and spare-parts inventory
- Improvement of operational safety
- Reduction of overall maintenance costs

2.4 Vibration-Based Condition Monitoring

Among the various condition-monitoring techniques—including thermal imaging, acoustic emission, oil analysis, and current-signature analysis—vibration monitoring is the most widely used for rotating machines. Vibration signals, typically measured using accelerometers, contain rich diagnostic information:

- Time-domain information: Amplitude variations, impulse patterns, and transient events.
- Frequency-domain information: Characteristic fault frequencies, harmonics, and sidebands.
- Time-frequency information: How the spectral content of the signal evolves over time.

2.5 Challenges with Traditional Approaches

Traditional vibration-based fault diagnosis follows a pipeline of signal acquisition, feature extraction, feature selection, and classification. While effective in controlled laboratory settings, this approach faces several challenges:

- Domain expertise: Manual feature extraction requires extensive knowledge of machine dynamics and signal processing.
- Feature engineering: Identifying the most informative features is often a trial-and-error process.
- Sensitivity to operating conditions: Handcrafted features may not generalise well across different speeds, loads, or machine types.
- Scalability: Designing features for every new machine or fault type is labour-intensive.

2.6 Motivation for Deep Learning

Deep learning addresses many of the limitations of traditional approaches by enabling automatic, hierarchical feature learning from data. Applied to vibration-based fault diagnosis, deep learning offers:

- Automatic extraction of relevant features from raw or minimally processed signals
- Ability to learn complex, non-linear relationships between vibration patterns and fault conditions
- Potential for improved generalisation through transfer learning and domain adaptation
- Scalability to multi-class, multi-fault diagnostic tasks

3. Problem Statement

Given a vibration signal collected from a rotating machine under various operating conditions, the problem is to develop a deep-learning-based system capable of:

1. Automatically processing the raw vibration signal through appropriate preprocessing and transformation steps.
2. Extracting discriminative features from both the time domain and the time-frequency domain representations of the vibration signal.
3. Classifying the machine's operating condition into one of several predefined categories (e.g., Normal, Bearing Fault, Unbalance, Misalignment, Gear Fault).
4. Optionally assessing the risk or severity of the diagnosed condition to support maintenance decision-making.

The system should minimise the need for manual feature engineering and should be designed to handle the inherent challenges of vibration data, including noise, non-stationarity, and variability across operating conditions.

4. Objectives

The specific objectives of this project are:

1. Acquire and process vibration signals from rotating machines, either from publicly available benchmark datasets or from laboratory measurements.
2. Remove noise from the raw vibration signals using appropriate digital filtering techniques.
3. Normalise the filtered signals to ensure consistent amplitude scaling across different recordings.
4. Segment the continuous vibration signals into fixed-length windows using a sliding-window approach.
5. Generate time-frequency representations (spectrograms) from the segmented windows using the STFT.
6. Extract time-domain features from the raw segmented windows using a 1D Convolutional Neural Network.
7. Extract time-frequency features from the spectrograms using a 2D Convolutional Neural Network.
8. Fuse the learned features from both CNN branches through concatenation.
9. Model temporal dependencies in the fused feature sequence using a BiLSTM network.
10. Classify the machine's operating condition using a fully connected layer followed by Softmax.
11. Evaluate the model's performance using appropriate classification metrics.
12. Optionally implement a fuzzy logic module for post-classification risk and severity assessment.

5. Proposed Methodology

The proposed system follows an end-to-end pipeline for vibration-based fault diagnosis:

Vibration Acquisition -> Preprocessing -> Segmentation -> Dual Representation -> CNN Feature Extraction -> Feature Fusion -> BiLSTM -> Softmax -> Fault Classification

The key design decisions of the proposed architecture are:

1. Dual-branch feature extraction: The architecture processes both the raw time-domain signal and its spectrogram representation in parallel, leveraging complementary information from both domains.
2. 1D CNN for raw vibration data: Applied directly to the raw vibration window to learn temporal patterns, impulse-like features, and amplitude-based characteristics.
3. 2D CNN for spectrograms: Applied to the STFT-derived spectrogram to learn patterns in the time-frequency plane.
4. Feature fusion: Feature vectors from both branches are concatenated to produce a unified representation.
5. BiLSTM for temporal modelling: Processes the fused features to capture temporal dependencies.
6. Softmax classification: Produces class probabilities for multi-class fault diagnosis.
7. Optional fuzzy logic: Provides interpretable risk/severity assessment.

This architecture is a proposed hybrid approach inspired by existing research on CNN-based vibration analysis, time-frequency representations, and recurrent networks. It is not claimed as an entirely novel invention; the contribution lies in the systematic integration of these established techniques into a single pipeline.

6. Proposed System Architecture

6.1 Main Architecture

The figure below illustrates the proposed hybrid deep learning architecture for vibration-based rotating-machine fault diagnosis.

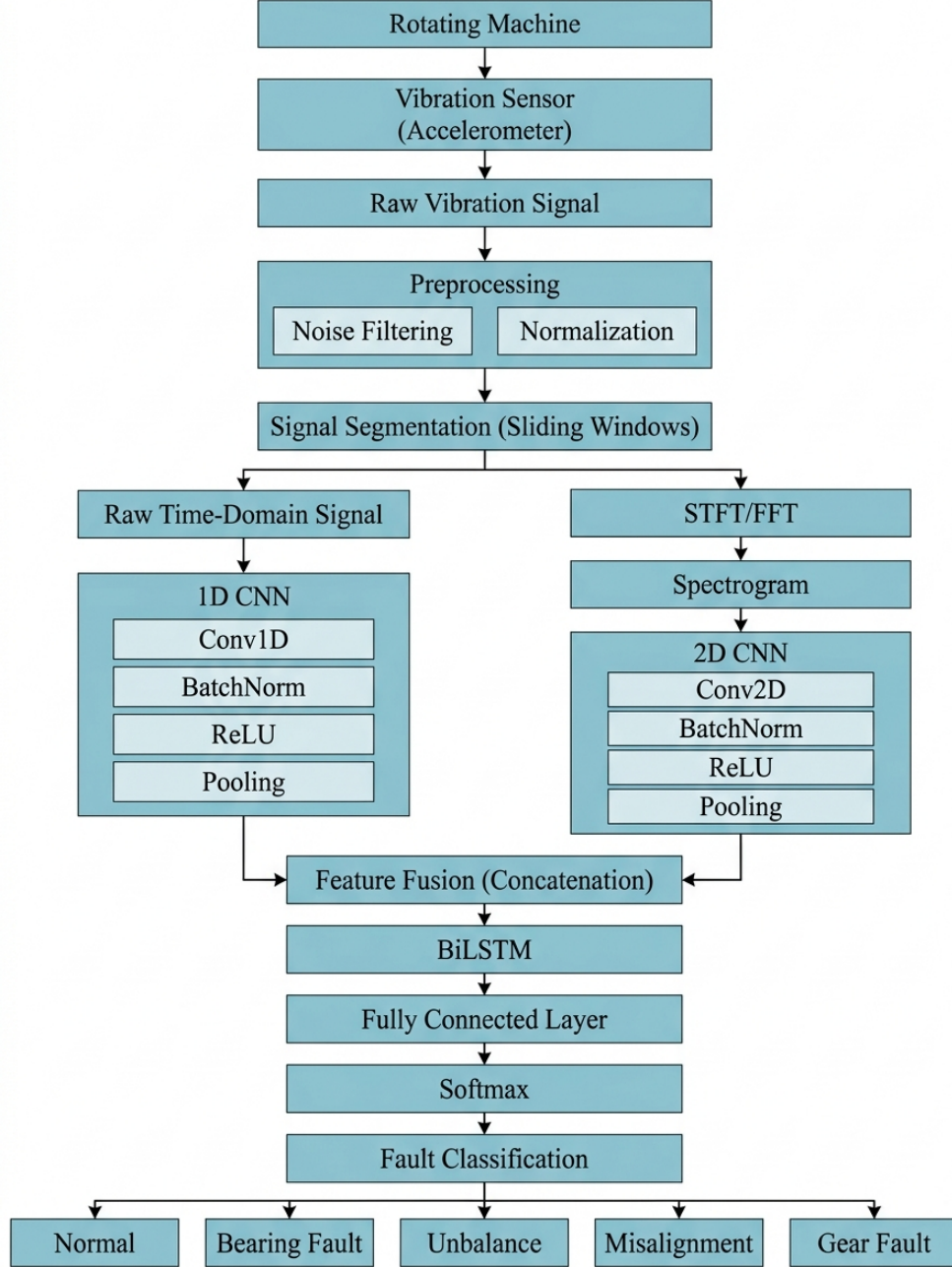


Figure 1. Proposed hybrid deep learning architecture for vibration-based rotating-machine fault diagnosis. The architecture employs a dual-branch feature-extraction strategy: a 1D CNN processes raw time-domain vibration windows while a 2D CNN processes STFT-derived spectrograms.

The architecture comprises the following major stages:

1. Data Acquisition: Vibration signals are collected from rotating machines using accelerometers.

2. Preprocessing: Noise filtering and normalisation are applied to the raw signals.
3. Segmentation: The preprocessed signal is divided into fixed-length windows using a sliding-window approach.
4. Dual-Branch Feature Extraction: The raw time-domain window feeds a 1D CNN; the STFT-derived spectrogram feeds a 2D CNN.
5. Feature Fusion: Feature vectors from both branches are concatenated.
6. Temporal Modelling: The fused features are processed by a BiLSTM layer.
7. Classification: A fully connected layer and Softmax activation produce fault-class probabilities.

6.2 Optional Fuzzy Logic Module

The figure below illustrates the optional fuzzy logic module for post-classification risk and severity assessment.

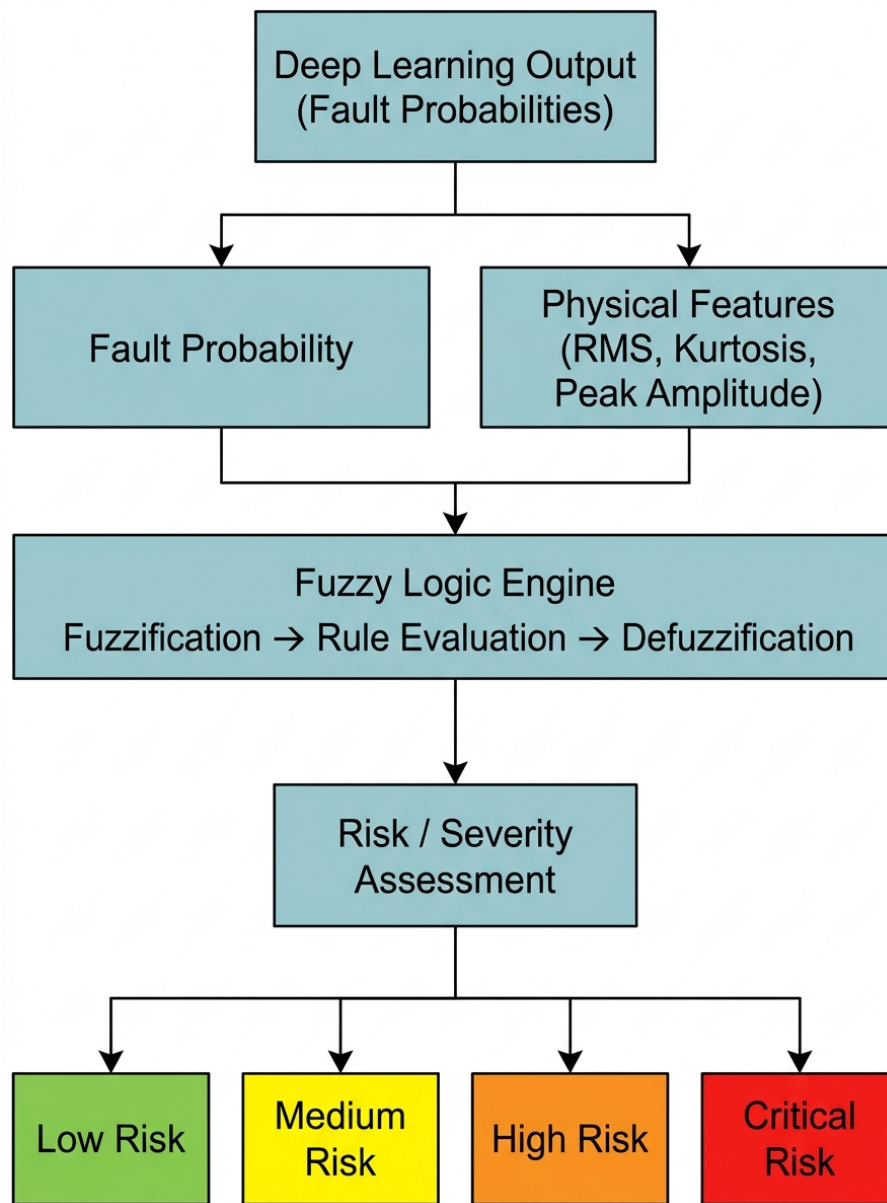


Figure 2. Optional fuzzy logic module for risk and severity assessment. The module combines deep-learning fault probabilities with physical signal features (RMS, Kurtosis, Peak Amplitude) through a fuzzy inference engine to produce interpretable risk levels.

7. Vibration Signal Acquisition

7.1 Accelerometers

Vibration signals from rotating machines are typically measured using accelerometers—transducers that convert mechanical vibration into electrical signals proportional to the acceleration of the vibrating surface. Piezoelectric accelerometers are the most commonly used type in industrial condition monitoring due to their wide frequency range, high sensitivity, and robustness.

7.2 Sensor Placement

Common mounting locations include:

- Bearing housings: For detecting bearing faults, inner-race defects, outer-race defects, and rolling-element faults.
- Motor housing: For detecting electrical and mechanical faults in motors.
- Gearbox casing: For detecting gear-tooth damage, wear, and mesh-frequency anomalies.

7.3 Sampling Frequency

The sampling frequency determines the temporal resolution and the maximum frequency that can be captured, according to the Nyquist–Shannon sampling theorem:

$$f_s \geq 2 * f_{\max}$$

For rotating-machine fault diagnosis, sampling frequencies typically range from a few kilohertz (e.g., 12 kHz) to tens of kilohertz (e.g., 48 kHz), depending on the machine's rotational speed and the fault-frequency range.

7.4 Time-Domain Vibration Signal

The output of an accelerometer, after analogue-to-digital conversion, is a discrete-time vibration signal:

$$x[n] = x(n * T_s), \quad n = 0, 1, 2, \dots, N-1$$

where $T_s = 1/f_s$ is the sampling period and N is the total number of samples.

8. Preprocessing

8.1 Noise Filtering

Industrial vibration signals are contaminated by noise from various sources. Common filtering techniques include:

- Low-pass filtering: Removes high-frequency noise above a specified cutoff frequency.
- Band-pass filtering: Retains only the frequency range of interest.
- Moving-average filtering: Smooths the signal by averaging consecutive samples.

A typical choice is a Butterworth band-pass filter with transfer function:

$$|H(jw)|^2 = 1 / (1 + (w/w_c)^{2n})$$

8.2 Normalisation

Z-score normalisation (standardisation):

$$\hat{x}[n] = (x[n] - \mu) / \sigma$$

Min-max normalisation:

$$\hat{x}[n] = (x[n] - x_{\min}) / (x_{\max} - x_{\min})$$

8.3 Signal Segmentation (Sliding Windows)

Continuous vibration signals are divided into shorter, fixed-length windows using a sliding-window approach. Define a window length L and a stride S :

$$w_k = x[k*S : k*S + L - 1], \quad k = 0, 1, 2, \dots$$

When $S < L$, consecutive windows overlap, increasing the number of samples and ensuring transient events near window boundaries are captured.

9. STFT and Spectrogram

9.1 Why FFT Alone May Not Be Sufficient

The FFT decomposes a signal into its constituent frequency components. However, the standard FFT assumes that the signal is stationary. Many vibration signals from faulty rotating machines are non-stationary; their spectral characteristics vary over time. The FFT provides an average spectral picture but does not reveal when specific frequency components are present.

9.2 The Short-Time Fourier Transform

The STFT applies the FFT to short, overlapping segments, providing a time-frequency representation:

$$\text{STFT}\{x[n]\}(m, w) = \sum x[n] * w[n-m] * e^{(-j*w*n)}$$

where $w[n]$ is a window function (e.g., Hanning, Hamming), m is the time index, and w is the angular frequency.

9.3 The Spectrogram

The spectrogram is the squared magnitude of the STFT:

$$S(m, w) = |\text{STFT}\{x[n]\}(m, w)|^2$$

It provides a 2D visual representation where:

- X-axis (horizontal): Time
- Y-axis (vertical): Frequency
- Colour intensity: Signal energy or amplitude at each time-frequency point

9.4 Why the Spectrogram Is Useful for Fault Diagnosis

- Captures both frequency content and its temporal evolution in a single 2D image.
- 2D CNNs can be directly applied to spectrogram images.
- Time-frequency patterns become computationally apparent in the spectrogram.

10. 1D CNN Branch

10.1 Purpose

The 1D CNN branch processes the raw, segmented time-domain vibration signal directly. It captures patterns such as amplitude modulations, impulse-like events, and waveform shapes characteristic of specific fault types.

10.2 Architecture

A typical 1D CNN branch consists of:

1. Conv1D: Applies learnable filters that slide along the time axis.
2. Batch Normalisation: Stabilises and accelerates training.
3. ReLU Activation: $\text{ReLU}(z) = \max(0, z)$. Introduces non-linearity.
4. Pooling: Reduces temporal dimensionality, retaining salient features.
5. Additional Conv1D blocks: Stacked for hierarchical feature learning.
6. Global Average Pooling: Produces a fixed-length feature vector.

10.3 What the 1D CNN Learns

Early layers learn low-level features such as edges, peaks, and oscillatory patterns. Deeper layers capture complex vibration patterns such as modulated impulse trains (bearing defects) or periodic amplitude variations (unbalance or misalignment).

11. 2D CNN Branch

11.1 Purpose

The 2D CNN branch processes the spectrogram representation, learning features from the two-dimensional time-frequency image that relate to how spectral content evolves over time.

11.2 Architecture

1. Conv2D: Applies 2D filters across both time and frequency axes.
2. Batch Normalisation: Normalises feature maps across the mini-batch.
3. ReLU Activation: Introduces non-linearity.
4. Max Pooling (2D): Reduces spatial dimensions.
5. Additional Conv2D blocks: Stacked for deeper feature learning.
6. Global Average Pooling (2D): Produces a fixed-length feature vector.

11.3 Why 2D CNN Is Appropriate for Spectrograms

Spectrograms are inherently two-dimensional images. Patterns of interest—harmonic lines, sidebands, impulse trains, energy concentrations—are spatial patterns in this image. 2D CNNs are well-suited to detect such patterns, similar to how they detect edges, textures, and objects in natural images.

12. Feature Fusion

12.1 Motivation

The two branches capture complementary aspects:

- 1D CNN -> time-domain features: Amplitude characteristics, temporal impulse patterns, oscillatory structures.
- 2D CNN -> time-frequency features: Spectral energy distributions, harmonic structures, spectral evolution.

By combining features from both branches, the model leverages a richer representation.

12.2 Concatenation

Feature fusion is performed through concatenation:

$$f_fused = [f_1D \parallel f_2D]$$

where f_1D is the feature vector from the 1D CNN (dimension $d1$), f_2D is from the 2D CNN (dimension $d2$), and f_fused has dimension $d1 + d2$.

13. BiLSTM

13.1 Why Temporal Dependencies Matter

Vibration signals are inherently sequential. Fault patterns may manifest as temporal sequences rather than instantaneous features. CNNs have limited receptive fields and do not explicitly model long-range dependencies.

13.2 Bidirectional LSTM

A BiLSTM processes the input sequence in both forward and backward directions. The outputs are concatenated:

$$h_t = [h_{t_forward} \parallel h_{t_backward}]$$

This captures context from both past and future elements, beneficial for recognising fault patterns that depend on both preceding and succeeding signal characteristics.

13.3 Design Rationale

The inclusion of BiLSTM is a design choice intended to capture temporal dependencies. Whether it provides measurable improvement is an empirical question to be validated through experimentation.

14. Softmax Classification

14.1 Fully Connected Layer

A fully connected layer maps the BiLSTM output to a vector of dimension C (number of fault classes):

$$z = W * h + b$$

14.2 Softmax Activation

The Softmax function converts logits into probabilities:

$$P(y=c | x) = \exp(z_c) / \sum_k \exp(z_k)$$

14.3 Example

Class	Probability
Normal	0.05
Bearing Fault	0.80
Unbalance	0.10
Misalignment	0.05

Predicted class = Bearing Fault (highest probability: 0.80)

15. Fault Classes

Potential fault classes (dependent on the selected dataset):

Class	Description
Normal	Healthy machine operation with no faults
Bearing Fault	Defects in inner race, outer race, or rolling elements
Unbalance	Uneven mass distribution causing excessive vibration
Misalignment	Angular or parallel offset between coupled shafts
Gear Fault	Tooth wear, pitting, or breakage in geared systems

Some datasets (e.g., CWRU Bearing Dataset) define more specific sub-classes with varying severity levels. The final class taxonomy should be clearly defined based on the chosen dataset.

16. Optional Fuzzy Logic Module

16.1 Motivation

The deep learning classifier answers: "What fault is most likely?"

The fuzzy logic module answers: "How severe or risky is the current condition?"

16.2 Distinction from the Deep Learning Classifier

The fuzzy logic module is NOT necessary for basic fault classification. It is an optional post-processing stage that adds interpretable risk assessment to the system's output.

16.3 Fuzzy Inputs

- Fault probability: The Softmax output from the deep learning classifier.
- RMS (Root Mean Square): Measure of overall signal energy.
- Kurtosis: Measure of signal "peakedness," sensitive to impulsive events.
- Peak amplitude: Maximum absolute amplitude of the signal.

16.4 Fuzzy Output

- Low Risk: Normal operation or negligible fault severity.
- Medium Risk: Early-stage fault; monitoring recommended.
- High Risk: Significant fault; maintenance should be scheduled.
- Critical Risk: Severe fault; immediate intervention required.

16.5 Example Fuzzy Rules

Rule	Condition	Conclusion
1	IF Fault Prob. LOW AND RMS LOW	Risk = LOW
2	IF Fault Prob. MEDIUM AND RMS ME	Risk = MEDIUM
3	IF Fault Prob. HIGH AND RMS HIGH	Risk = CRITICAL
4	IF Kurtosis HIGH AND Fault Prob. MEI	Risk = HIGH
5	IF Peak Amp. HIGH AND RMS MEDIU	Risk = HIGH
6	IF Fault Prob. LOW AND Kurtosis LOW	Risk = LOW

16.6 Fuzzy Inference Process

1. Fuzzification: Converts crisp input values into degrees of membership in fuzzy sets using predefined membership functions.
2. Rule Evaluation: Evaluates all fuzzy rules against the fuzzified inputs.
3. Defuzzification: Combines activated rule outputs into a single crisp risk level (e.g., centroid or mean-of-maximum method).

17. Training Procedure

17.1 Dataset Preparation

Publicly available benchmark datasets include:

- CWRU Bearing Dataset: Vibration signals at 12 kHz and 48 kHz under various fault conditions and loads.
- Paderborn University Bearing Dataset: Vibration and current signals with artificial and real-world faults.
- MFPT Dataset: Vibration signals from bearings under normal and faulty conditions.

17.2 Data Splitting

- Training set (60-70%): Used to train model parameters.
- Validation set (15-20%): Used to monitor performance and tune hyperparameters.
- Test set (15-20%): Used to evaluate final model performance on unseen data.

17.3 Avoiding Data Leakage

Data splitting should be performed at the recording level (entire recordings assigned to one subset) rather than at the window level, to prevent information leakage and artificially inflated performance metrics.

17.4 Loss Function

Categorical cross-entropy loss for multi-class classification:

$$L = -\sum_c y_c * \log(y_{\text{hat}_c})$$

17.5 Optimiser

Adam (Adaptive Moment Estimation) optimiser, combining momentum-based and adaptive learning-rate methods.

17.6 Early Stopping

Monitors validation loss and halts training if no improvement for a specified number of consecutive epochs. The model weights corresponding to the lowest validation loss are retained.

17.7 Hyperparameter Tuning

Key hyperparameters include:

- Learning rate, batch size, number/size of convolutional filters
- Number of BiLSTM hidden units, dropout rate
- Window length and stride for segmentation
- STFT parameters (window size, hop length, FFT points)

18. Evaluation Metrics

18.1 Accuracy

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

Accuracy can be misleading when classes are imbalanced. A model that always predicts the majority class achieves high accuracy without learning fault patterns.

18.2 Precision

$$\text{Precision}_c = \text{TP}_c / (\text{TP}_c + \text{FP}_c)$$

Measures the proportion of correct positive predictions. High precision means few false positives.

18.3 Recall (Sensitivity)

$$\text{Recall}_c = \text{TP}_c / (\text{TP}_c + \text{FN}_c)$$

Measures the proportion of actual positives correctly identified. Critical for fault classes where missed faults have serious consequences.

18.4 F1-Score

$$\text{F1}_c = 2 * \text{Precision}_c * \text{Recall}_c / (\text{Precision}_c + \text{Recall}_c)$$

The harmonic mean of precision and recall, providing a balanced performance measure.

18.5 Confusion Matrix

A detailed breakdown of predictions versus true labels for all classes. Useful for identifying which fault types the model confuses with each other.

18.6 Handling Class Imbalance

- Macro-averaged metrics: Average metrics equally across classes.
- Weighted-averaged metrics: Account for class frequencies.
- Techniques: Oversampling (SMOTE), undersampling, or class-weighted loss functions.

19. Expected Results

The expected outcomes, to be confirmed through experimentation:

1. Correct identification of normal and faulty conditions with high classification accuracy.
2. Improved feature representation through dual-branch extraction combining time-domain and time-frequency information.
3. Ability to capture temporal patterns through BiLSTM, potentially enhancing fault pattern recognition.
4. Potentially improved robustness compared to single-representation approaches.
5. Interpretable risk assessment if fuzzy logic module is implemented.

Note: Specific accuracy percentages are not reported, as these depend on the dataset, experimental configuration, and hyperparameter settings. All quantitative results will be determined through rigorous experimentation.

20. Advantages

1. Automatic feature learning: Eliminates the need for manual, domain-expert-driven feature engineering.
2. Complementary information: Dual-branch design captures both temporal and spectro-temporal patterns.
3. Temporal modelling: BiLSTM captures sequential dependencies and temporal context.
4. Multi-class diagnosis: Softmax supports simultaneous classification into multiple fault categories.
5. Extensibility to severity assessment: Optional fuzzy logic extends from classification to risk assessment.
6. End-to-end trainability: Joint optimisation of all feature-extraction and classification parameters.

21. Limitations

1. Requirement for sufficient labelled data, especially for rare fault conditions.
2. Computational cost of the dual-branch CNN + BiLSTM architecture.
3. Sensitivity to operating conditions not represented in training data.
4. Dataset dependency: Performance tied to the specific training dataset.
5. Generalisation challenges: May not transfer directly to unseen machines without fine-tuning.
6. Class imbalance: Potential bias toward majority classes.
7. Sensor noise and environmental variability may affect performance.

22. Future Scope

1. Transfer learning: Pre-training on large vibration datasets and fine-tuning for specific machines.

2. Domain adaptation: Adapting models across different machines or operating conditions.
3. Transformer architectures: Vision Transformers for spectrograms, temporal Transformers for sequence modelling.
4. Explainable AI (XAI): Grad-CAM, SHAP, attention visualisation for model interpretability.
5. Real-time edge deployment: Model compression (pruning, quantisation, knowledge distillation).
6. Remaining Useful Life (RUL) prediction: Extending from classification to prognostics.
7. Multi-sensor fusion: Incorporating accelerometers, current sensors, temperature sensors, acoustic sensors.
8. Online condition monitoring: Streaming pipeline for real-time fault alerts and trending.

23. Conclusion

This report has presented a proposed hybrid deep learning architecture for the fault diagnosis of rotating machines using vibration signals. The architecture systematically integrates several established techniques into a coherent end-to-end diagnostic pipeline:

- Vibration signal acquisition and preprocessing ensure clean, consistently scaled input data.
- Signal segmentation using sliding windows produces individual samples for neural-network input.
- Dual-branch feature extraction using a 1D CNN (raw time-domain) and a 2D CNN (STFT spectrograms) captures complementary diagnostic information.
- Feature fusion through concatenation combines both representations into a unified feature vector.
- A BiLSTM layer models temporal dependencies within the fused feature sequence.
- A fully connected layer and Softmax activation produce probabilistic fault-class predictions.
- An optional fuzzy logic module provides interpretable risk and severity assessment.

The contribution of this work lies not in claiming entirely novel components, but in the thoughtful integration of these proven techniques into a single, end-to-end pipeline for rotating-machine fault diagnosis. Future work will focus on experimental validation using benchmark vibration datasets, comparison with baseline methods, and exploration of extensions such as transfer learning, Transformer architectures, explainable AI, and real-time edge deployment.

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25. Appendix: Module List and Implementation Workflow

A. List of Modules

Module	Name	Description
M1	Vibration Acquisition	Data collection from accelerometers
M2	Noise Filtering	Noise removal using band-pass filtering
M3	Normalisation	Z-score or min-max normalisation
M4	Signal Segmentation	Sliding-window segmentation
M5	STFT / Spectrogram	Time-frequency transformation
M6	1D CNN Extraction	Features from raw time-domain windows
M7	2D CNN Extraction	Features from spectrogram images
M8	Feature Fusion	Concatenation of CNN feature vectors
M9	BiLSTM Modelling	Temporal dependency modelling
M10	FC Layer	Dense layer for classification mapping
M11	Softmax	Probabilistic fault classification
M12	Fuzzy Logic	Optional risk/severity assessment

B. Implementation Workflow

Phase 1: Data Preparation

- Dataset selection and acquisition
- Exploratory data analysis
- Noise filtering
- Signal normalisation
- Sliding-window segmentation and labelling

Phase 2: Feature Representation

- STFT computation and spectrogram generation
- Data formatting for 1D and 2D inputs

Phase 3: Model Development

- 1D CNN branch design and implementation
- 2D CNN branch design and implementation
- Feature-fusion layer implementation
- BiLSTM layer integration
- Fully connected and Softmax output layer
- End-to-end model assembly

Phase 4: Training and Validation

- Train/validation/test split (recording-level)

- Model training with Adam optimiser
- Early stopping and learning-rate scheduling
- Hyperparameter tuning on validation set
- Best-model selection

Phase 5: Evaluation

- Test-set evaluation
- Accuracy, precision, recall, F1-score computation
- Confusion matrix analysis
- Comparison with baseline methods

Phase 6: Optional Fuzzy Logic

- Define fuzzy variables and membership functions
- Define fuzzy rules
- Implement fuzzy inference engine
- Integrate with deep learning output

Phase 7: Documentation and Presentation

- Report preparation
- Architecture diagrams
- Viva preparation